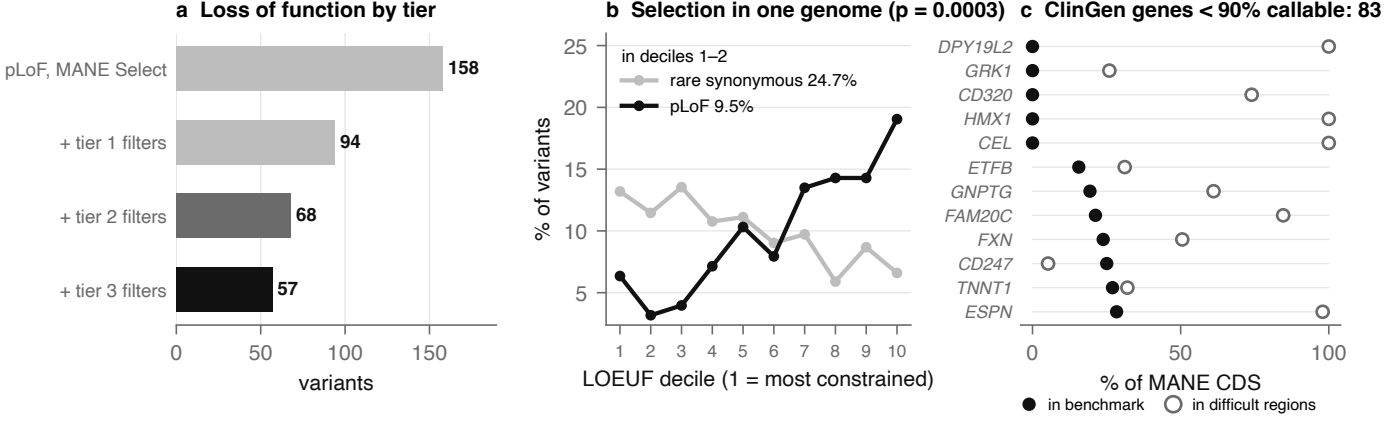
a Loss of function by tier
pLoF, MANE Select
+ tier 1 filters
+ tier 2 filters
+ tier 3 filters
158
94
68
57
0
50
100
150
variants
b Selection in one genome (p = 0.0003)
in deciles 1–2
rare synonymous 24.7%
pLoF 9.5%
% of variants
25
20
15
10
5
1
2
3
4
5
6
7
8
9
10
LOEUF decile (1 = most constrained)
c ClinGen genes < 90% callable: 83
DPY19L2
GRK1
CD320
HMX1
CEL
ETFB
GNPTG
FAM20C
FXN
CD247
TNNT1
ESPN
0
50
100
% of MANE CDS
in benchmark
in difficult regions